

Karthik.K

Data Engineer | Python Developer

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Profile Summary

Data Engineer and Python Developer with hands-on experience in building ETL pipelines, data migration, database integration, and large-scale data processing using Python, SQL, MSSQL, MySQL, MongoDB, and ClickHouse. Alongside data engineering, contributing to AI-based speech processing projects by developing ASR evaluation pipelines, transcript processing workflows, and speech model performance analysis. Strong foundation in backend and full-stack development, with a focus on building scalable, efficient, and maintainable enterprise data and AI solutions.

Technical Skills

Programming Languages: Python, SQL, JavaScript

Data Engineering: ETL Development, Data Migration, Data Integration, Data Transformation, Data Validation, Batch Processing, Data Modeling

AI & Speech Processing: Automatic Speech Recognition (ASR), Speech Data Processing, Transcript Processing, Model Evaluation, Data Preprocessing, pandas, scikit-learn

Front-End Development : HTML5, CSS3, JavaScript, React.js, Bootstrap

Back-End Development: Node.js, Express.js

Database Management: MySQL, MSSQL, MongoDB, ClickHouse

Tools & Technologies: Git, GitHub, Visual Studio Code, Docker

Professional Experience

Software Engineering Intern | Wyzmindz Solutions Pvt. Ltd.

May 2026 – Present

- Developed a Python-based ETL job automation framework that monitored FTP locations for incoming files and automatically triggered Talend and Python jobs for unattended processing.
- Deployed the automation as a Windows Service using NSSM to ensure reliable background execution, service recovery, and continuous job monitoring.
- Built data migration and validation workflows across MSSQL, MySQL, MongoDB, and ClickHouse to support enterprise data processing requirements.
- Contributed to an AI-based speech processing initiative by generating MGT datasets, comparing ASR model outputs, and applying SWAR, NAWAR, and ASCAR evaluation algorithms.
- Implemented logging, error handling, and execution tracking to improve reliability, traceability, and troubleshooting of production workflows.

Full Stack Development Intern | Sourcesys Technologies

Feb 2026 – Apr 2026

- Developed responsive web interfaces using HTML, CSS, JavaScript, and React.js.
- Built REST APIs and backend functionalities using Node.js and Express.js.
- Integrated MongoDB for data storage, retrieval, and application management.
- Worked on authentication, API integration, debugging, testing, and feature enhancement as part of the full-stack development workflow.

Projects

ETL Process Automation Using NSSM

Designed and implemented an event-driven ETL automation framework using Python that continuously listened for incoming files on FTP servers and automatically orchestrated Talend and Python data processing jobs. The framework was deployed as a Windows Service using NSSM with integrated logging, monitoring, and service recovery mechanisms to provide reliable, fault-tolerant, and unattended execution of production ETL workflows.

AI Speech Processing & MGT Dataset Generation Platform

Designed and developed a speech evaluation pipeline to generate Machine Ground Truth (MGT) datasets by comparing transcript outputs from multiple Automatic Speech Recognition (ASR) models. Implemented SWAR, NAWAR, and ASCAR evaluation algorithms to assess semantic accuracy, named entity recognition, and contextual similarity, enabling the creation of high-quality transcripts for speech model training and performance benchmarking.

Chat Application (MERN Stack)

[Live link](#)

Developed a full-stack real-time chat and collaborative whiteboard application using the MERN stack with JWT-based authentication, Socket.IO for instant messaging, RESTful APIs, and MongoDB for data management. Deployed the application on Vercel and Render.

AI-Based Sentiment Analysis System for Product Reviews

[Live-Link](#)

Developed an end-to-end sentiment analysis system using Python, TF-IDF, and Naive Bayes to classify product reviews. Built an interactive Streamlit web application supporting single review input, bulk CSV upload, audio review analysis, dashboard visualization, and automated PDF report generation.

Education

Bachelor of Technology in Information Technology

2022-2026

Er.Perumal Manimekalai College Of Engineering.

CGPA: 9.5/10

Higher Secondary Education (+2)

2020-2022

SRI VIJAY VIDHYALAYA HIGHER SECONDARY SCHOOL, HOSUR.

Certifications and Achievements

Deloitte Australia Technology Job Simulation on Forage - June 2025

[Certificate](#)

The Complete Full-Stack Web Development Bootcamp (Udemy) (Sept 2025)

[Certificate](#)

TCS iON Career Edge – Young Professional (Jul 2025)

[Certificate](#)

IBM SkillsBuild – Artificial Intelligence Fundamentals (Oct 2025)

[Certificate](#)